

Chapter 11 One-Pager — Coalition Formation and Detection

Research on the possibilities for applying Artificial Intelligence in computer games

Alexander Andreev

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Problem. Chapters 2-10 all leaned on one crutch: a two-player game with an exact best-response and exploitability oracle, so “did it work?” was one number. Add a third player and alliances become possible — form, exploit, betray — while Nash turns both intractable and guarantee-free, since it ignores coalitions entirely. Chapter 11 removes the crutch deliberately. This is the thesis frontier, not a consolidation step.

Approach. A native 4-player **So Long Sucker** engine — the bargaining game of Hausner, Nash, Shapley and Shubik, published in 1964 — implemented from the published rules, with a 2-player **minimax endgame** as its only exact anchor and the De Carufel & Jerade endgame analysis as the (not yet cross-checked) reference. On top: a **coalition detector** (help/harm from chip placement), **Shapley credit** redefined as the value of a member’s win probability, a **coalition-aware MAPPO** trainer blending coalition and winner-takes-all reward by a weight **alpha**, and **EGTA + spinning-top** analysis reused from Chapter 10. **All numbers below are measured**, with one caveat: **scale_results.json** is a **pre-fix** run cited only as evidence of the bug below — the authoritative figures are **sweep_scale.json** (5 seeds) and **post-fix smoke_results.json**.

Key results (measured).

- *Certified wherever exactness exists.* 100/100 random games terminate, all rewards are zero-sum, **endgame_mismatches: 0** against exact minimax. The detector recovers a **planted {0,1} alliance** from the move stream alone (pair score **10.0**, cross-pair entries 0 or -1; a sanity check on one scripted, overt pair), and Shapley reproduces the glove and majority toys exactly, core emptiness included.
- *The chapter’s biggest finding is a bug, and its lesson outlived it.* Two red checks shared one cause: **~99.5% of random games in this engine end in deadlock**, so the winner falls to **_most_chips**, whose lowest-index tie-break gave seat 0 twice its fair share (all-random winners [94,42,33,31]). With an unbiased tie-break, symmetric Shapley spread fell **0.54 -> 0.013** and winners went uniform; the same fix deflated an impressive **0.87** hero win rate to an honest **~0.41** against 0.25 for random play. In a game that nearly always ends near-tied, the tie-break rule is load-bearing.
- *“No effect at scale” was overturned by finding the right knob — with a caveat.* **alpha** dominates and the **0.3 default sits in a dead zone**: at **alpha ~ 0** the proxy-credit gap is **+0.0376 +/- 0.0103** (the Shapley agents’ score is ~4.5x the sparse **0.0109**), while **every scale-tier alpha >= 0.3 cell is negative**. The effect is larger in the scale tier (12x the smoke tier at the same synergy; game size and training length change together), and the cheap critic-value proxy beats the rollout credit (**+0.038 vs +0.013**). But as implemented, neither credit encodes who helped whom, and the coalition score also counts hostile interactions: a dense reward raises the score, which is not yet evidence of learned coalitions.
- *The coalition score is bought, not free.* **alpha = 0** drops hero win rate to **~0.29**, close to random play (0.25), while **alpha >= 0.1** holds **~0.52**.
- *Strongly cyclic, honestly short of the bar.* The skill-ladder pool is transitive-dominant (cyclic **0.25-0.31**), but a coalition pool pushes cyclic to **~0.57-0.69** — large, yet still failing the strict “>50% dominance” rule, so the check stays red. Harness: **4/5 PASS**.

Thesis connection. Contribution #1 generalises cleanly: opponent modelling lifts from “what kind of player is this” to “who is allied with whom”, read straight off the moves. Contribution #2 gets its problem statement sharpened rather than solved — with no minimax value and an **empty core**, “safe” cannot mean bounded deviation from equilibrium; a behavioural anchor in the spirit of piKL is a candidate, not a guarantee. Contribution #3 gains a working substitute for exploitability in EGTA plus the spinning-top decomposition, inheriting Chapter 10’s caveat that the population you assemble decides whether you see a ladder or a wheel.

Open questions. Reconciling the engine’s turn and tie-break model against De Carufel & Jerade is the highest-value follow-up — both the seat-0 bug and the red cyclic check trace to documented simplifications, and the paper’s theorems are not yet checked. Whether a 3- or 4-player-aware EGTA projection surfaces the cycling the 2-type collapse discards. Whether a credit built on the detector’s help/harm matrix, tested against a coalition-free dense reward, produces real coalitions. And what “safe” can mean at all when the core is empty.